Module 15 – Cloud Fundamentals

Introduction to Cloud Computing

What is Cloud Computing?

Cloud computing is the delivery of computing resources such as servers, storage, databases, networking, and software over the internet on a pay-as-you-go basis. It allows users to access computing resources without owning physical infrastructure.

Traditional IT Deployment vs Cloud Computing

Traditional IT	**Cloud Computing**
Physical servers	Virtual resources
High initial investment	Pay only for usage
Manual hardware maintenance	Managed by cloud provider
Limited scalability	Easy and quick scaling
Longer deployment time	Rapid deployment

Virtualization

Virtualization is the process of creating virtual versions of physical resources such as servers, storage, and operating systems. It allows multiple virtual machines to run on a single physical server, improving resource utilization and reducing costs.

Service-Oriented Architecture (SOA)

SOA is a software architecture where applications are built using independent services that communicate over a network. It promotes reusability, flexibility, and easier maintenance.

Cloud vs On-Premises

Cloud Advantages

- Lower infrastructure cost
- High scalability
- Automatic updates
- High availability
- Global accessibility

On-Premises Advantages

- Full control over infrastructure
- Better for strict security and compliance
- No dependency on internet connectivity

Cloud Service Models

Infrastructure as a Service (IaaS)

Provides virtual servers, storage, and networking.

Example: Amazon EC2

Platform as a Service (PaaS)

Provides a platform to develop, test, and deploy applications.

Example: AWS Elastic Beanstalk

Software as a Service (SaaS)

Provides software applications over the internet.

Examples: Gmail, Microsoft 365

Cloud Deployment Models

- **Public Cloud:** Services shared over the internet (AWS, Azure).
- **Private Cloud:** Dedicated cloud for one organization.
- **Hybrid Cloud:** Combination of public and private cloud.
- **Community Cloud:** Shared by organizations with common requirements.

Major Cloud Providers

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

Amazon EC2 – Instances, AMIs, Security Groups & Key Pairs

What is EC2?

Amazon EC2 (Elastic Compute Cloud) provides scalable virtual servers in the cloud.

Launching an EC2 Instance

- Choose an Amazon Machine Image (AMI)

- Select an instance type
- Configure networking
- Add storage
- Configure security groups
- Create/select a key pair
- Launch the instance

Amazon Machine Image (AMI)

An AMI is a template containing the operating system and software required to launch an EC2 instance.

Types of AMIs

- Public AMIs
- Private AMIs
- Marketplace AMIs

EC2 Instance Types

Different configurations based on CPU, memory, and storage.

Example: **t2.micro** (Free Tier eligible)

Security Groups

Security Groups act as virtual firewalls controlling inbound and outbound traffic to EC2 instances.

Key Pairs

Key pairs (.pem files) provide secure authentication when connecting to EC2 instances through SSH.

Amazon ECS – Container Basics

What is a Container?

A container packages an application along with all its dependencies, ensuring consistent execution across environments.

What is Amazon ECS?

Amazon Elastic Container Service (ECS) is a fully managed container orchestration service for running Docker containers.

ECS vs EC2

ECS	EC2
Runs containers	Runs virtual machines
Managed container service	Virtual server

Creating and Managing Containers

- Create task definitions
- Create clusters
- Deploy containers
- Monitor running services

Amazon S3 – Buckets, Objects & Storage Classes

What is Amazon S3?

Amazon Simple Storage Service (S3) is an object storage service used to store files and backups.

Buckets and Objects

- **Bucket:** Container that stores objects.
- **Object:** Actual file stored inside a bucket.

Creating an S3 Bucket

- Create bucket
- Select region
- Configure permissions
- Upload files

Uploading & Downloading Objects

Files can be uploaded or downloaded using the AWS Console, CLI, or SDK.

Bucket Permissions

- Public
- Private

Storage Classes

- Standard
- Intelligent-Tiering
- Standard-IA
- Glacier

Additional Features

- Lifecycle Policies
- Versioning

Amazon EBS – Block Storage

What is Amazon EBS?

Elastic Block Store (EBS) provides persistent block-level storage for EC2 instances.

Creating and Attaching Volumes

Volumes can be created separately and attached to running EC2 instances.

EBS Volume Types

- gp2
- gp3
- io1

Snapshots

Snapshots are backups of EBS volumes stored in Amazon S3.

Amazon VPC – Networking

What is Amazon VPC?

Amazon Virtual Private Cloud (VPC) is an isolated virtual network within AWS.

Public and Private Subnets

Public Subnet

- Connected to the Internet Gateway
- Internet accessible

Private Subnet

- No direct internet access
- Used for databases and internal services

Route Tables

Route tables define how network traffic moves between subnets and gateways.

Internet Gateway (IGW)

Provides internet access to resources in public subnets.

NAT Gateway

Allows private subnet resources to access the internet without being publicly accessible.

Security Groups

Act as instance-level firewalls.

VPC Peering

Allows communication between two VPCs.

Elastic Load Balancer (ELB)

What is ELB?

Elastic Load Balancer distributes incoming traffic across multiple EC2 instances to improve availability.

Application Load Balancer (ALB)

- Layer 7
- HTTP/HTTPS
- Path-based routing

Network Load Balancer (NLB)

- Layer 4
- TCP/UDP
- High-performance workloads

ALB vs NLB

ALB	NLB
Layer 7	Layer 4
HTTP/HTTPS	TCP/UDP
Path-based routing	High-speed networking

Target Groups

Target groups contain the EC2 instances that receive traffic.

Health Checks

Regularly monitor instance health before forwarding requests.

Amazon RDS

What is Amazon RDS?

Amazon Relational Database Service (RDS) is a managed relational database service.

Managed vs Self-Managed Database

Managed (RDS)

- AWS handles backups, updates, and maintenance.

Self-Managed

- User manages everything.

Supported Database Engines

- MySQL
- PostgreSQL
- SQL Server
- Amazon Aurora

Multi-AZ Deployment

Creates a standby database in another Availability Zone for automatic failover.

Backups

- Automated backups
- Manual snapshots

Amazon DynamoDB

What is DynamoDB?

Amazon DynamoDB is a fully managed NoSQL database service.

NoSQL Database

Stores key-value and document data instead of relational tables.

RDS vs DynamoDB

RDS	**DynamoDB**
SQL	NoSQL
Fixed schema	Flexible schema
Best for relational data	Best for high-scale applications

Creating a Table

Specify:

- Partition Key

- Sort Key (optional)

Query and Scan

- Query retrieves specific data.
- Scan searches the entire table.

Use Cases

Ideal for applications requiring high performance and low latency.

AWS Lambda

What is Serverless Computing?

Serverless computing allows developers to run code without managing servers.

AWS Lambda

Executes code automatically when triggered by events.

Event Sources

- Amazon S3
- API Gateway
- DynamoDB Streams

Creating a Lambda Function

- Select runtime
- Write/upload code
- Configure triggers
- Deploy

Supported Runtimes

- Java
- Python
- Node.js

Pricing

Pay only for execution time and resources consumed.

Integration

Can integrate with S3, API Gateway, DynamoDB, SNS, and many other AWS services.

AWS API Gateway

What is API Gateway?

AWS API Gateway is a managed service used to create, publish, secure, and manage APIs.

Creating a REST API

- Create API
- Add resources
- Define methods
- Deploy API

HTTP Methods

- GET
- POST
- PUT
- DELETE

Lambda Integration

API Gateway can invoke Lambda functions to create serverless APIs.

Deployment Stages

- Development
- Staging
- Production

Security Features

- Throttling
- API Keys
- IAM Authorization
- Basic authentication controls

Summary

Cloud computing provides on-demand access to IT resources over the internet with high scalability, flexibility, and cost efficiency. AWS offers services such as **EC2** for virtual machines, **ECS** for containerized applications, **S3** and **EBS** for storage, **VPC** for networking, **ELB** for load balancing, **RDS** and **DynamoDB** for databases, **Lambda** for serverless computing, and **API Gateway** for API management. Together, these services help build secure, scalable, highly available, and modern cloud applications.